

The WKB Approximation

The WKB (Wentzel–Kramers–Brillouin) approximation is often referred to as the semi-classical approximation to solve Schrödinger equation. It allows us to take a classical limit of our system while also capturing the leading quantum behavior of particles. It is convenient because it recovers the classical idea of trajectories and momenta for particles moving through a slowly-varying potential, while also providing corrections to this classical picture. As opposed to the variational method, the WKB approximation will allow us to approximate all kinds of solutions to Schrödinger's equation, not just the ground state.

Preliminaries and Classical Limit

We start with the Schrödinger equation in three dimensions with some generic potential $V(\mathbf{x}, t)$:

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t) + V(\mathbf{x}, t) \psi(\mathbf{x}, t). \quad (1)$$

As you have done in your homework, we can write the wavefunction without any loss of generality in terms of an amplitude and a phase as

$$\psi(\mathbf{x}, t) = \sqrt{\rho(\mathbf{x}, t)} e^{\frac{i}{\hbar} S(\mathbf{x}, t)}, \quad (2)$$

where $\rho(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2$ is the probability density and $S(\mathbf{x}, t)$ is some phase. Both are real functions. Recall that with this parametrization we found the probability density current to be

$$\mathbf{J}(\mathbf{x}, t) = \frac{\hbar}{2mi} \left(\psi^*(\mathbf{x}, t) \nabla \psi(\mathbf{x}, t) - \psi(\mathbf{x}, t) \nabla \psi^*(\mathbf{x}, t) \right) = \rho(\mathbf{x}, t) \frac{\nabla S(\mathbf{x}, t)}{m}. \quad (3)$$

The interpretation was that $\nabla S(\mathbf{x}, t)$ is analogous to the momentum (or $\nabla S(\mathbf{x}, t)/m$ is analogous to the velocity, $p = mv$) of a classical particle at position $\mathbf{x}$ and time t . It is in this sense that we will think of the phase of the wavefunction as encoding information about the dynamics of our particle.

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To see this, take a free particle traveling through space with a wavefunction

$$\psi(\mathbf{x}, t) = C e^{\frac{i}{\hbar} (\mathbf{p} \cdot \mathbf{x} - Et)}, \quad (4)$$

where C is a normalization constant, $\mathbf{p}$ is the particle three-momentum and $E = \mathbf{p}^2/2m$ is its energy. Clearly, for this simplest case,

$$S(\mathbf{x}, t) = \mathbf{p} \cdot \mathbf{x} - Et \implies \begin{cases} \nabla S(\mathbf{x}, t) = \mathbf{p}, \\ \frac{\partial S(\mathbf{x}, t)}{\partial t} = -E. \end{cases} \quad (5)$$

As expected, we recover the momentum, but what is more, we also learn that the time derivative is just $-E$. Both of these equalities are consistent with the classical action in classical mechanics. Note that for this single particle case $\rho(\mathbf{x}, t) = |C|^2$, which is subtle since a free particle is not normalizable in all space, but that is a whole separate issue.

What is important here is that in this example, the phase depends trivially on position and time thanks to the absence of a potential. Once you turn on a potential $V(\mathbf{x}, t)$, the phase $S(\mathbf{x}, t)$ will in general be more complicated and could display more interesting dynamics than the motion of our free particle plane wave. It is precisely this dynamical problem that the WKB approximation aims to solve. But before delving into the approximation, let us see a classical limit emerging from Schrödinger's equation.

Plugging the form of Eq. (2) into Schrödinger's equation, we find

$$i\hbar \left(\frac{1}{2} \frac{1}{\sqrt{\rho}} \frac{\partial \rho}{\partial t} + \frac{i}{\hbar} \sqrt{\rho} \frac{\partial S}{\partial t} \right) e^{\frac{i}{\hbar} S} = -\frac{\hbar^2}{2m} \left[\nabla^2 (\sqrt{\rho} e^{\frac{i}{\hbar} S}) \right] + V \sqrt{\rho} e^{\frac{i}{\hbar} S}, \quad (6)$$

where I dropped my arguments for simplicity. The Laplacian term is

$$\nabla^2 (\sqrt{\rho} e^{\frac{i}{\hbar} S}) = e^{\frac{i}{\hbar} S} \left[\nabla^2 \sqrt{\rho} + \frac{2i}{\hbar} \nabla \sqrt{\rho} \cdot \nabla S + \frac{i}{\hbar} \sqrt{\rho} \nabla^2 S - \frac{1}{\hbar^2} \sqrt{\rho} (\nabla S)^2 \right]. \quad (7)$$

Dividing both sides by $e^{\frac{i}{\hbar} S}$ and separating real and imaginary parts (we can do this because we know that both S and ρ are real functions), you should find:

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} + V - \frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} = 0, \quad (8)$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \frac{\nabla S}{m} \right) = \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0. \quad (9)$$

The second equation is just the continuity equation for probability density that we have already seen (this is significant: the conservation of probability is encoded in the imaginary part of the Schrödinger equation). The first equation is the one that concerns us and the WKB method.



Eq. (7) looks very similar to the Hamilton–Jacobi equation from classical mechanics, which reads (see Eq. 2.8.18 of Shankar's book):

$$\frac{\partial S}{\partial t} + H(\mathbf{x}, \nabla S, t) = 0, \quad (10)$$

where $H(\mathbf{x}, \mathbf{p}, t)$ is the Hamiltonian of the system expressed in terms of coordinates $\mathbf{x}$ and momenta $\mathbf{p}$. For a single non-relativistic particle in a potential $V(\mathbf{x}, t)$, the Hamiltonian is $H = \mathbf{p}^2/2m + V(\mathbf{x}, t)$, so using our analogy between momentum and the gradient of the phase, the Hamilton–Jacobi equation becomes

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} + V(\mathbf{x}, t) = 0. \quad (11)$$

It is intriguing that following this reasoning, we see that Eq. (7) reduces to the Hamilton–Jacobi equation only in the limit $\hbar \rightarrow 0$ (classical limit), where the last term vanishes. This is one way to see how classical behavior emerges from quantum mechanics in the appropriate limit.

First Hints of Semi-Classical Behavior

As we will see below, the WKB approximation is technically an expansion in $\hbar$. One obvious feature of the last term in Eq. (7) is that it comes in at order $\hbar^2$, and so it should be small when the WKB method is useful. This last term is sometimes called the *quantum potential*:

$$Q(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho(\mathbf{x}, t)}}{\sqrt{\rho(\mathbf{x}, t)}}. \quad (11)$$

Physically, it encodes quantum effects in “wave mechanics”, and it is responsible for quantum phenomena such as tunneling and interference. While called a potential, this is very different from $V(\mathbf{x}, t)$ as it depends on our state $\psi(\mathbf{x}, t)$ itself (through $\rho(\mathbf{x}, t)$). Without it, we recover the Hamilton–Jacobi equation.

Let us see this more explicitly by taking the gradient of Eq. (7) and identifying $\mathbf{v} = \nabla S/m$ as before. We get a $F = ma$ type equation, with $F = -\nabla(V + Q)$:

$$m \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla(V + Q). \quad (12)$$

In one dimension, we can interpret the term inside the parenthesis on the left-hand side as just $dv/dt = \partial_t v + v \partial_x v$. In the WKB approximation, the phase $S(\mathbf{x}, t)$ of the wavefunction satisfies the classical Hamilton–Jacobi equation to a good approximation, and the dynamics of the particle can be understood in terms of classical trajectories.

The WKB Approximation (In Practice)

Okay, so we understand that there is some limit where the quantum behavior is irrelevant when $\hbar \rightarrow 0$. But in the way we wrote it above, it is hard to see what we are approximating and hard to see how to use this in the real world. It still does not make our jobs easier to solve Schrödinger’s equation and we never really found any expansion parameter ($\hbar$ is a constant after all!).

To see the usefulness of the WKB approximation, let us consider the time-independent Schrödinger equation in three dimensions with some generic potential $V(\mathbf{x})$ and energy eigenvalue E :

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}) + V(\mathbf{x}) \psi(\mathbf{x}) = E \psi(\mathbf{x}). \quad (13)$$

Note that the fact that we are seeing the energy eigenvalue E means that our wavefunction $\psi(\mathbf{x})$ is an energy eigenstate (that is, we have used $\hat{H}|\psi\rangle = E|\psi\rangle$ to arrive at the equation above). This also means we are dealing with a stationary state, so the time dependence of the wavefunction is just a global phase $e^{-iEt/\hbar}$ that we can factor out and ignore for the purposes of this discussion.

Because we will be thinking about the dynamics of a particle with a given momentum, it is also convenient to re-write Schrödinger’s equation in terms of the wavevector $\mathbf{k}$, defined as $\mathbf{p} = \hbar \mathbf{k}$. Rearranging, we have

$$\hat{p}^2 \psi(\mathbf{x}) = 2m(E - V(\mathbf{x})) \psi(\mathbf{x}) \iff -\hbar^2 \nabla^2 \psi(\mathbf{x}) = 2m(E - V(\mathbf{x})) \psi(\mathbf{x}), \quad (14)$$

where again I emphasize that $\hat{p} = -i\hbar \nabla$ is the momentum operator that acts on functions as opposed to $\hat{p}|\psi\rangle$ which is the equivalent operator acting on kets in Hilbert space. Let me rewrite this in a more suggestive way by defining the local momentum $\mathbf{p}(\mathbf{x})$ as

$$p(\mathbf{x}) \equiv \sqrt{2m(E - V(\mathbf{x}))} \implies -\hbar^2 \nabla^2 \psi(\mathbf{x}) = p(\mathbf{x})^2 \psi(\mathbf{x}). \quad (15)$$

Now, instead of working with Eq. (2), let us make a similar, but conceptually distinct ansatz for the energy eigenstate wavefunction:

$$\psi(\mathbf{x}) = e^{\frac{i}{\hbar} S(\mathbf{x})} \quad (\text{with energy } E), \quad (16)$$

where $S(\mathbf{x})$ is now a complex function rather than a real one. Note the lack of a prefactor to the exponential, which is now not necessarily a complex exponential. This form is as general as Eq. (8).¹

Plugging this form into the time-independent Schrödinger equation in Eq. (13) (where now we have the extra assumption of this being an energy eigenstate and only have to worry about the phase $S(\mathbf{x})$), we find

$$-\hbar^2 \nabla^2 \left(e^{\frac{i}{\hbar} S(\mathbf{x})} \right) = p(\mathbf{x})^2 e^{\frac{i}{\hbar} S(\mathbf{x})} \implies \boxed{-i\hbar \nabla^2 S(\mathbf{x}) + (\nabla S(\mathbf{x}))^2 = p(\mathbf{x})^2}. \quad (17)$$

¹We can always write $S(\mathbf{x}) = R(\mathbf{x}) + iI(\mathbf{x})$ and then $\psi(\mathbf{x}) = e^{iR(\mathbf{x})/\hbar} e^{-I(\mathbf{x})/\hbar}$, which is equivalent to Eq. (8) as long as we interpret $\sqrt{\rho(\mathbf{x})} = e^{-I(\mathbf{x})/\hbar}$.

Now here comes the key step in the WKB approximation: we expand $S(\mathbf{x})$ in powers of $\hbar$ as

$$S(\mathbf{x}) = S_0(\mathbf{x}) + \hbar S_1(\mathbf{x}) + \hbar^2 S_2(\mathbf{x}) + \dots, \quad (18)$$

where $S_0(\mathbf{x})$ is the leading order term, $S_1(\mathbf{x})$ is the first-order correction, and so on.

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This should make you uncomfortable for two reasons:

1. $\hbar$ is a dimensionful constant in nature, so what good can it do as an expansion parameter? In natural units, $\hbar = 1$.
2. How can we be sure that this series converges without already knowing the full solution?

The first concern is resolved by noting that what really matters is the ratio of $\hbar$ to some characteristic action S_c in the problem, which has the same units as $\hbar$ (action has units of energy times time, or equivalently momentum times length). Thus, in some sense, the expansion parameter is really a dimensionless ratio like $\hbar/S_c$. The second concern is more subtle and to the best of knowledge, there is no rigorous proof of convergence. That being said, we know for a fact that taking $\hbar$ to be small recovers classical behavior, so we have a good reason to assume that the series is at least asymptotic, meaning that it provides a good approximation to the true solution for sufficiently small $\hbar$.

Plugging this expansion into Eq. (14) and collecting terms order by order in $\hbar$, we find:

$$\mathcal{O}(\hbar^0) : (\nabla S_0(\mathbf{x}))^2 = p(\mathbf{x})^2, \quad (19)$$

$$\mathcal{O}(\hbar^1) : i\nabla^2 S_0(\mathbf{x}) - 2\nabla S_0(\mathbf{x}) \cdot \nabla S_1(\mathbf{x}) = 0, \quad (20)$$

$$\mathcal{O}(\hbar^2) : i\nabla^2 S_1(\mathbf{x}) - 2\nabla S_0(\mathbf{x}) \cdot \nabla S_2(\mathbf{x}) - (\nabla S_1(\mathbf{x}))^2 = 0, \quad (21)$$

$$\vdots \quad (22)$$

The leading order equation is just the Hamilton–Jacobi equation, which we have already seen emerges in the classical limit. In that sense, truncating the series at leading order gives us the classical behavior of the system. From now on, things can get hairy, so let us work in 1D for simplicity:

$$\mathbf{x} \rightarrow x, \quad \nabla \rightarrow \frac{d}{dx}, \quad p(\mathbf{x}) \rightarrow p(x), \quad S_n(\mathbf{x}) \rightarrow S_n(x). \quad (23)$$

Let us solve the leading order equation first. We have

$$|\nabla S_0(\mathbf{x})| = p(\mathbf{x}) \implies \nabla S_0(x) = \pm p(x) \quad (\text{in 1D}), \quad (24)$$

which can be integrated to give

$$S_0(x) = \pm \int_{x_0}^x p(x') dx' = \pm \int_{x_0}^x \sqrt{2m(E - V(x'))} dx'. \quad (25)$$

where the initial point x_0 is arbitrary and the integral is taken along some path from x_0 to x (the point x_0 just determines the constant of integration, which in turn is just a trivial overall phase for the wavefunction.) The choice of the $\pm$ sign corresponds to the two possible directions of motion for the particle. This is the leading order WKB solution, which captures the classical behavior of the particle in the potential $V(x)$. For some region with a constant potential, we can write the solution as

$$\psi(x) \approx e^{\frac{i}{\hbar} S_0(x)} = e^{\pm \frac{i}{\hbar} p(x-x_0)}, \quad (26)$$

with $p = \sqrt{2m(E - V)}$ being the local momentum of the particle in that region. This is a rather simple form for a solution. If we know how to integrate the local momentum, we can write down an approximate solution to Schrödinger's equation without having to solve any differential equations.

The next order correction $S_1(x)$ obeys its own differential equation. Using the solution for $S_0(x)$, we can write it as

$$\frac{dS_1(x)}{dx} = \frac{i}{2} \frac{\frac{d^2 S_0(x)}{dx^2}}{\frac{dS_0(x)}{dx}} = \frac{i}{2} \frac{dp(x)}{p(x)}. \quad (27)$$

Integrating this, we find the first-order correction to the WKB solution:

$$S_1(x) = \frac{i}{2} \int_{x_0}^x \frac{dp(x')}{p(x')} dx' = \frac{i}{2} \int_{x_0}^x \frac{d}{dx'} (\ln p(x')) dx' = \frac{i}{2} \ln \frac{p(x)}{p(x_0)}. \quad (28)$$

Note that the initial point x_0 is again arbitrary and just determines the constant of integration. In principle, it has nothing to do with x_0 from the leading order solution. Finally, using both solutions, we have a better approximation for $S(x) \simeq S_0(x) + \hbar S_1(x)$, which gives us the WKB wavefunction

$$\psi(x) \simeq e^{\frac{i}{\hbar}(S_0(x) + \hbar S_1(x))} = e^{\frac{i}{\hbar} S_0(x)} e^{i S_1(x)} = \sqrt{\frac{p(x_0)}{p(x)}} e^{\pm \frac{i}{\hbar} \int_{x_0}^x p(x') dx'}. \quad (29)$$

There are two arbitrary constants of integration, x_0 and x_0 . A more convenient form of the WKB solution is given when we have an initial condition for $\psi(x)$ at some point x_0 , which allows us to fix the constant of integration x_0 in terms of x_0 and write the solution as

$$\psi(x) \simeq \psi(x_0) \sqrt{\frac{p(x_0)}{p(x)}} e^{\pm \frac{i}{\hbar} \int_{x_0}^x p(x') dx'}, \quad (30)$$

which is the most standard form of the WKB solution to the time-independent Schrödinger equation. The particle behaves like a plane wave with a local momentum $p(x)$ that depends on the position through the potential $V(x)$. The amplitude of the wavefunction is modulated by the factor $1/\sqrt{p(x)}$, which accounts for the fact that the probability density must be conserved as the particle moves through regions of different potential.

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We can check that our expansion is valid by asking that $|\hbar S_1(x)| \ll |S_0(x)|$. This ought to be true for small steps in x away from x_0 . Taking a derivative with respect to x , we find the condition

$$\hbar \left| \frac{dp(x)}{dx} \frac{1}{p(x)} \right| \ll p(x). \quad (31)$$

Rearranging, we find the WKB validity condition:

$$\left| \frac{d\lambda(x)}{dx} \right| \ll 1, \quad (32)$$

where $\lambda(x) = 2\pi\hbar/p(x)$ is the local de Broglie wavelength of the particle. This condition states that the WKB approximation is valid when the de Broglie wavelength of the particle changes slowly compared to the scale of the wavelength itself. Physically, this means that the potential $V(x)$ must vary slowly over distances comparable to the wavelength of the particle.

Example: Potential Step

Let us now apply the WKB approximation to a simple problem,

$$\psi(x) \simeq \psi(x_0) \sqrt{\frac{p(x_0)}{p(x)}} e^{\pm \frac{i}{\hbar} \int_{x_0}^x p(x') dx'}, \quad (33)$$

where $p(x) = \sqrt{2m(E - V(x))}$ is the local momentum of the particle at position x . Make the potential now a step function, such that $V(x) = 0$ for $x < 0$ and $V(x) = V_0$ for $x \geq 0$, where V_0 is a positive constant. In the region $x < 0$, the local momentum is $p(x) = \sqrt{2mE}$, and the WKB solution is trivial:

$$\psi(x) \simeq \psi(x_0) e^{\pm \frac{i}{\hbar} \sqrt{2mE}(x-x_0)}. \quad (34)$$

In the region $x \geq 0$, the local momentum is $p(x) = \sqrt{2m(E - V_0)}$, and the WKB solution is

$$\psi(x) \simeq \psi(x_0) \sqrt{\frac{\sqrt{2mE}}{\sqrt{2m(E - V_0)}}} e^{\pm \frac{i}{\hbar} \int_{x_0}^x \sqrt{2m(E - V_0)} dx'} = \psi(x_0) \sqrt{\frac{E}{E - V_0}} e^{\pm \frac{i}{\hbar} \sqrt{2m(E - V_0)}(x-x_0)}. \quad (35)$$

Clearly, at the boundary $x = 0$, the WKB solution is not valid because the potential changes abruptly, and the local momentum $p(x)$ becomes a delta function at that point (recall our validity condition for the WKB approximation in Eq. (32)).



This is a known limitation of the WKB approximation: it fails near turning points where the local momentum goes to zero, and it also fails at points where the potential changes abruptly. Nevertheless, away from these points, the WKB approximation provides a good description of the particle's behavior in the potential. In particular, it captures the oscillatory behavior of the wavefunction in classically allowed regions (where $E > V(x)$) and the exponential decay in classically forbidden regions (where $E < V(x)$), which are key features of quantum mechanics that are not captured by classical mechanics.

Example: The Harmonic Oscillator

Now let us consider the harmonic oscillator potential $V(x) = \frac{1}{2}m\omega^2 x^2$, where ω is the angular frequency of the oscillator. The local momentum is $p(x) = \sqrt{2m(E - \frac{1}{2}m\omega^2 x^2)}$, and the WKB solution is

$$\psi(x) \simeq \psi(x_0) \sqrt{\frac{\sqrt{2mE}}{\sqrt{2m(E - \frac{1}{2}m\omega^2 x^2)}}} e^{\pm \frac{i}{\hbar} \int_{x_0}^x \sqrt{2m(E - \frac{1}{2}m\omega^2 x'^2)} dx'}. \quad (36)$$

This integral is not trivial, but in principle we have a closed-form expression for the WKB solution in terms of the integral of the local momentum.



At this point, you can read Sakurai's discussion on WKB. They will tell you how to prove the WKB quantization condition, from which we can derive the energy eigenvalues of bound states in a potential.

The WKB quantization condition states that the integral of the local momentum over one complete cycle of the motion must be equal to an integer multiple of $2\pi\hbar$:

$$\oint p(x) dx = 2\pi\hbar \left(n + \frac{1}{2} \right), \quad (37)$$

where n is a non-negative integer (the quantum number).



You can understand this quantization condition physically: in the classical limit, the particle cannot be found outside in the region where $E < V(x)$, so it is confined to a finite region of space. Its phase, therefore, must be single-valued after one complete cycle of the motion, which leads to a quantization condition. This argument is somewhat modified by the fact that the first quantum correction $S_1(x)$ also contributes to the phase of the wavefunction, and it does allow the wavefunction to leak into the classically forbidden region, but by a fixed amount. That is the origin of the $1/2$ in the quantization condition, which is a purely quantum effect that has no classical analog. This analysis gives different quantization conditions for different potentials. The one above is for a potential that can be linearly approximated near the turning points (that is, that can be expanded as $V(x) \approx V(x_{\text{turn}}) + V'(x_{\text{turn}})(x - x_{\text{turn}})$), which is the case for the harmonic oscillator. For an infinite square well, the quantization condition is $\oint p(x)dx = 2\pi\hbar(n + 1)$ instead, because the potential is infinite at the boundaries and the wavefunction must vanish there no matter what (the potential certainly cannot be linearly approximated near the turning points in this case, so the $1/2$ correction does not apply).

For the harmonic oscillator, we have

$$\oint p(x)dx = 2 \int_{-\sqrt{\frac{2E}{m\omega^2}}}^{\sqrt{\frac{2E}{m\omega^2}}} \sqrt{2m \left(E - \frac{1}{2}m\omega^2 x^2 \right)} dx. \quad (38)$$

Evaluating this integral, we find

$$\oint p(x)dx = 2\pi E/\omega. \quad (39)$$

Applying the WKB quantization condition, we get

$$2\pi E/\omega = 2\pi\hbar \left(n + \frac{1}{2} \right) \implies E = \hbar\omega \left(n + \frac{1}{2} \right). \quad (40)$$

This is the exact energy spectrum of the quantum harmonic oscillator, which shows that the WKB approximation can give us the correct energy eigenvalues for this system, even though it is an approximation method.

It turns out that for other bound states, the WKB approximation gives us a better approximation for the energy eigenvalues for higher quantum numbers n , and it becomes exact in the limit of large n (this is classical limit after all!). So, the WKB method is complementary to the variational method: while the variational method is best suited for approximating the ground state energy, the WKB approximation can be used to approximate excited states and can provide insights into the behavior of the wavefunction in different regions of space. That being said, it is somewhat messy for 3D problems. For instance, in Hydrogen, the potential is spherically symmetric, so it is convenient to work in spherical coordinates and separate variables. The WKB approximation can be applied to the radial part of the wavefunction, but it is not as straightforward as in 1D because for higher energy levels, the radial wavefunction has multiple nodes and the potential has a centrifugal barrier that complicates the calculation.